Research Paper Roadmap for Task-Oriented Multimodal IR Agents

1. Problem Definition

- **Research Focus:** The goal is to develop a task-oriented multimodal information retrieval (IR) agent that uses multiple modalities (text, image, audio, video) to retrieve information and complete tasks based on user input.
- **Key Objective:** The agent should not only retrieve information but also execute actions, such as answering questions, making decisions, or performing tasks based on a combination of data types (e.g., text, image, audio).
- **Contextual Challenge:** How can agents combine various modalities to complete user-defined tasks in real-time? And how can they reason over multimodal inputs to perform tasks efficiently?

2. Core Research Questions

- How do multimodal inputs (text, image, audio) interact in a task-oriented system?
- What models or frameworks can be applied to reason over multimodal data and generate actionable outputs from it?
- How can task completion be effectively modeled and evaluated in multimodal search systems?
- What existing technologies (e.g., NLP, computer vision, voice recognition) can be integrated into this agent?
- How can a retrieval agent choose the most relevant modality for a task or query?
- What are the performance metrics to evaluate task-oriented agents with multimodal inputs?

3. Literature Review

- **Task-Oriented Dialogue Systems:** Review existing literature on task-oriented dialogue agents (e.g., Rasa, DialogFlow, BERT-based agents), and identify gaps in combining them with multimodal inputs.
- **Multimodal Search:** Look into multimodal IR (e.g., CLIP, VisualBERT, and UNITER) and how they retrieve content based on text + image/audio combinations.
- **Multimodal Reasoning:** Explore research on multimodal reasoning in AI, such as neural symbolic machines, graph neural networks, and knowledge graphs for multimodal data.

- **Task Completion in AI:** Investigate agents that complete tasks rather than just retrieving information (e.g., assistants like Siri, Alexa).
- **Evaluation Metrics for Multimodal Agents:** Investigate evaluation methods used in multimodal systems and task-oriented models to understand how task completion and retrieval performance are measured.

4. Methodology

1. **System Architecture:**
   - **Agent Design:** Design the agent as a modular system where each modality (text, image, audio) has its own processing pipeline. The agent should process each modality independently and then combine them to understand the task.
     - **Text Processing**: Use NLP models like BERT or GPT for task understanding.
     - **Image Processing**: Use models like CLIP, ResNet, or ViT for image recognition.
     - **Audio Processing**: Leverage speech-to-text and sound recognition models (e.g., DeepSpeech or Wav2Vec2).
     - **Task Logic**: Combine outputs from each modality and reason over the gathered information.
   - **Task-oriented Pipeline:** Build a task pipeline where, depending on the query, the agent chooses which modality (or combination) to use. This decision-making can be done through a task classifier or rule-based system.
2. **Task Completion Model:**
   - Use **reinforcement learning (RL) or supervised learning** for training the system to complete tasks based on multimodal inputs. For example, you can create a reward function based on task success (e.g., correct answers, successful execution of actions).
3. **Multimodal Fusion:**
   - Implement **multimodal fusion** techniques, such as:
     - **Concatenation Fusion:** Simple concatenation of feature vectors from each modality.
     - **Attention Fusion:** Using attention mechanisms like cross-modal attention to focus on the most relevant parts of each modality.

- **Late Fusion:** Combining results from each modality at the decision level.

4. **Evaluation Strategy:**

   - Develop a **multi-metric evaluation framework** to assess the agent's task completion and retrieval effectiveness. Metrics might include:

     - Precision/Recall/F1-Score for task success rate
     - Task Completion Time
     - User Satisfaction (measured by subjective evaluation)
     - Cross-modal relevance (how well the agent integrated multimodal data to complete the task)

5. **Dataset & Tools:**

   - **Datasets:**
     - MS COCO, Visual Question Answering (VQA), or Flicker30k for multimodal image-text tasks.
     - LibriSpeech or Common Voice for audio-to-text tasks.
     - MultiWOZ or Taskmaster for building task-oriented dialogues and interactive agents.
   - **Tools/Frameworks:**
     - Transformers (Hugging Face) for text models (e.g., BERT, GPT).
     - CLIP, ResNet, or VGG for image processing.
     - DeepSpeech, Wav2Vec2 for audio processing.
     - PyTorch/TensorFlow for model implementation.
     - Rasa or DialogFlow for the task-oriented dialogue framework.

8-Week Roadmap for Task-Oriented Multimodal IR Agents

(Information Processing &Management (IP & M) Target Journal Standards (Q1 venue))

Week 1: Literature Review & Research Question Definition

- **Objective:**
  - Conduct a **comprehensive literature review** focusing on:
    - Task-Oriented Dialogue Systems: Explore existing work on multimodal agents, chatbots, and dialogue systems (e.g., Rasa, BERT-based agents).
    - Multimodal Information Retrieval: Review multimodal fusion models (e.g., CLIP, VisualBERT, VQA).
    - Task Completion in AI: Discuss systems that complete tasks using multimodal input (e.g., AI assistants, robots).
    - Fusion Strategies: Examine how multiple modalities are integrated (e.g., attention mechanisms, fusion strategies).
    - Evaluation Metrics: Identify evaluation frameworks for task-oriented IR systems, including precision, recall, task success rate.
  - Define your **research problem**, **novel contributions**, and **hypotheses**.
- **Deliverables:**
  - **Literature review (2-3 pages)**: Clearly identify research gaps and relate them to your study.
  - **Research questions:** Ensure that they align with IP&M's focus on IR and task-oriented AI systems.
  - **Hypotheses:** Define hypotheses related to multimodal task completion and fusion methods.

Week 2: System Design & Methodology

- **Objective:**
  - **Design the system architecture** for the **task-oriented multimodal IR agent**:
    - Define the components for text, image, and audio processing.
    - Describe how these modalities will interact to achieve task completion.
    - Implement the fusion strategies (e.g., concatenation, cross-modal attention, late fusion).
  - Develop the **evaluation framework** for assessing both **multimodal retrieval effectiveness** and **task completion success** (e.g., task time, success rate, and efficiency).
  - Start drafting the **methodology section** for the paper, focusing on clear and reproducible experimental designs.
- **Deliverables:**
  - **System architecture**: Detailed block diagram of the system.
  - **Methodology section (1-2 pages)**: Describe how multimodal data is processed, fused, and utilized for task completion.
  - **Evaluation strategy:** Define key metrics such as precision, recall, F1 score, and task success rate.

Week 3: Data Collection & Preprocessing

- **Objective:**
  - Collect appropriate **multimodal datasets** for training and evaluation:
    - **MS COCO**, **VQA**, **Flickr30k** for image-text tasks.
    - **LibriSpeech**, **Common Voice** for speech-to-text tasks.
    - Explore existing **task-oriented datasets** for dialogue-based agents (e.g., **MultiWOZ** or **Taskmaster**).
  - **Preprocess the data:**
    - **Text**: Tokenization and embedding (e.g., **BERT**, **GPT**).

        - **Image**: Feature extraction (e.g., **CLIP**, **ResNet**).
        - **Audio**: Use models like **DeepSpeech** or **Wav2Vec2** for speech-to-text conversion.
    - Ensure **data quality** and create **training-validation-test splits**.
- **Deliverables:**
    - **Preprocessed datasets** ready for model training.
    - Write a section on **data sources and preprocessing** for the methodology.

Week 4: Model Implementation (Text, Image, Audio Processing)

- **Objective:**
    - Implement models for each modality:
        - **Text**: Fine-tune **BERT** or **GPT** models for text understanding.
        - **Image**: Implement **CLIP** or **ResNet** for image recognition and understanding.
        - **Audio**: Use **Wav2Vec2** or **DeepSpeech** for **speech-to-text** conversion.
    - Begin integrating the models into a unified **task completion pipeline**.
    - The pipeline should **allow the agent to decide which modality (or combination) to use depending on the task and query.**
- **Deliverables:**
    - Implemented models for **text**, **image**, and **audio** modalities.
    - Integrated **task-oriented multimodal pipeline**.
    - Draft a section on **model implementation** for the paper.

Week 5: Multimodal Fusion & Task Completion Logic

- **Objective:**
  - Implement the **fusion strategies** (e.g., **concatenation**, **cross-modal attention**, **late fusion**).
  - Develop the **task logic**, allowing the agent to determine how to use the fused modalities to **complete tasks**.
  - Test the agent's **ability to handle multimodal inputs** (e.g., a query combining text, image, and audio) and successfully complete tasks (e.g., providing a detailed response or executing actions).
- **Deliverables:**
  - Implemented **fusion techniques** and **task completion logic**.
  - **Initial testing** of the agent's ability to complete multimodal tasks.
  - **Update methodology section**: Include descriptions of **fusion techniques** and **task completion logic**.

Week 6: Evaluation & Refinement

- **Objective:**
  - Conduct **evaluation experiments** using the defined metrics (e.g., **task success rate**, **precision**, **recall**).
  - Perform **ablation studies** to evaluate the impact of different **fusion strategies** and **task completion methods**.
  - Refine the system based on **experimental feedback** and **optimization** results.
  - Write the **evaluation results section**, presenting both quantitative results (graphs, tables) and qualitative assessments (e.g., user study or scenario-based testing).
- **Deliverables:**
  - **Evaluation results** with detailed analysis (graphs, tables, and narrative).
  - **Ablation study results** comparing different fusion and task strategies.
  - **Updated paper draft** with **evaluation section**.

Week 7: Finalize Results & Write the Paper

- **Objective:**
  - Finalize **results**, ensuring clarity and completeness in presenting all findings.
  - Write the **discussion** section, focusing on the implications of your results and how they contribute to the field.
  - Finalize **conclusion**, highlighting **novel contributions**, potential real-world applications, and **future work**.
  - Refine and **format the paper** according to **IP&M submission guidelines**.
- **Deliverables:**
  - **Completed paper draft** (Introduction, Related Work, Methodology, Results, Discussion, Conclusion).
  - Final paper **formatting according to journal guidelines** (e.g., font, citations, structure).

Week 8: Paper Submission

- **Objective:**
  - Perform a **final review** of the paper to ensure it meets the highest standards of clarity, accuracy, and conciseness.
  - **Submit the paper** to **IP&M**, ensuring all submission requirements are met (e.g., manuscript format, figures, references).
  - Prepare for potential **post-submission revisions** or **feedback** from reviewers.
- **Deliverables:**
  - **Final paper submission** to **IP&M**.
  - Ready for any **revisions** based on reviewer **feedback**.

Tips for Ensuring Successful Submission to IP&M:

1. **Clarity & Structure**: Ensure that the paper is well-organized and clearly explains both the technical aspects (methodology, model details) and practical applications (real-world use cases).
2. **Reproducibility**: Consider sharing your code and datasets (if permissible) to increase reproducibility and transparency, which IP&M highly values.
3. **Impactful Evaluation**: Provide robust empirical results, including user studies or task completion scenarios, that demonstrate the real-world applicability of your system.
4. **Quick Revisions**: If IP&M asks for revisions, be prepared to address reviewer feedback quickly, as their review cycle is relatively fast.

Conclusion:

Task-Oriented Multimodal IR Agents present an exciting challenge and will make a significant contribution to the AI field. With the proper methodology, resources, and clear research questions, it's a feasible project that can culminate in a high-quality paper with both theoretical and practical relevance.